

YARN: EFFICIENT CONTEXT WINDOW EXTENSION OF LARGE LANGUAGE MODELS

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ABSTRACT

Rotary Position Embeddings (RoPE) have been shown to effectively encode positional information in transformer-based language models. However, these models fail to generalize past the sequence length they were trained on. We present YaRN (Yet another RoPE extensioN method), a compute-efficient method to extend the context window of such models, requiring 10x less tokens and 2.5x less training steps than previous methods. Using YaRN, we show that LLaMA models can effectively utilize and extrapolate to context lengths much longer than their original pre-training would allow, while also surpassing previous the state-of-the-art at context window extension. In addition, we demonstrate that YaRN exhibits the capability to extrapolate beyond the limited context of a fine-tuning dataset. Code is available at <https://github.com/jquesnelle/yarn>.

1 INTRODUCTION

Transformer-based Large Language Models (Vaswani et al., 2017) (LLMs) have become the near-ubiquitous choice for many natural language processing (NLP) tasks where long-range abilities such as *in-context learning* (ICL) has been crucial.

In performing the NLP tasks, the maximal length of the sequences (the *context window*) determined by its training processes has been one of the major limits of a pretrained LLM. Being able to dynamically extend the context window via a small amount of fine-tuning (or without fine-tuning) has become more and more desirable. To this end, the position encodings of transformers are the center of the discussions.

The original Transformer architecture used an absolute sinusoidal position encoding, which was later improved to a learnable absolute position encoding (Gehring et al., 2017). Since then, relative positional encoding schemes (Shaw et al., 2018) have further increased the performance of Transformers. Currently, the most popular relative positional encodings are *T5 Relative Bias* (Roberts et al., 2019), *RoPE* (Su et al., 2022), *XPos* (Sun et al., 2022), and *ALiBi* (Press et al., 2022).

One reoccurring limitation with positional encodings is the inability to generalize past the context window seen during training. While some methods such as ALiBi are able to do limited generalization, none are able to generalize to sequences significantly longer than their pre-trained length (Kazemnejad et al., 2023).

Some works have been done to overcome such limitation. (Chen et al., 2023) and concurrently (kaio-kendev, 2023) proposed to extend the context length by slightly modifying RoPE via Position Interpolation (PI) and fine-tuning on a small amount of data. As an alternative, (bloc97, 2023a) proposed the "NTK-aware" interpolation by taking the loss of high frequency into account. Since then, two improvements of the "NTK-aware" interpolation have been proposed, with different emphasis:

- the "Dynamic NTK" interpolation method (emozilla, 2023) for pre-trained models without fine-tuning.
- the "NTK-by-parts" interpolation method (bloc97, 2023b) which performs the best when fine-tuned on a small amount of longer-context data.

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Let $\theta = \text{diag}(\theta_1, \dots, \theta_{|D|/2})$. In complex coordinates, we define

$$f_{\mathbf{W}}(\mathbf{x}_m, m, \theta) = e^{im\theta} \mathbf{W} \mathbf{x}_m, \quad (5)$$

for any linear operator $\mathbf{W}$. The functions f_q, f_k in RoPE are given by

$$f_q = f_{\mathbf{W}_q}, f_k = f_{\mathbf{W}_k}. \quad (6)$$

where $\theta_d = b^{-2d/|D|}$ and $b = 10000$. This way, RoPE associates each (complex-valued) hidden neuron with a separate frequency θ_d . The benefit of doing so is that the dot product between the query vector and the key vector only depends on the relative distance $m - n$.

In later discussions, a context length interpolation usually aims to modify the equation Eq. 5. To set up a uniform convention for these discussions, note that a modification $f'_{\mathbf{W}}$ can take the following form:

$$f'_{\mathbf{W}}(\mathbf{x}_m, m, \theta) = f_{\mathbf{W}}(\mathbf{x}_m, g(m), \mathbf{h}(\theta)), \quad (7)$$

where $g(m)$ is a map between real numbers and $\mathbf{h}(\theta)$ acts on the entries of the diagonal matrix θ uniformly by $\text{diag}(h(\theta_1), \dots, h(\theta_{|D|/2}))$ according to a function h . g and h are method-dependent functions.

In the subsequent sections, when we introduce a new interpolation method of the form Eq. 7, we only specify the functions $g(m)$ and $h(\theta_d)$.

2.2 ADDITIONAL NOTATIONS

Given the pretrained maximal context length L , our goal is to extend it to $L' > L$ either with or without finetuning. We introduce the notion of *scale factor* s defined by $s = \frac{L'}{L}$.

For the convenience of some discussions, we also introduce *wavelength* λ_d associated with the d -th hidden dimension of RoPE as follows:

$$\lambda_d = \frac{2\pi}{\theta_d} = 2\pi b^{\frac{2d}{|D|}}. \quad (8)$$

The wavelength describes the length of tokens needed in order for the rotary position embedding at dimension d to perform a full rotation (2π).

2.3 RELATED WORK

Position Interpolation (PI) is one of the earlier works extending context lengths of RoPE proposed by Chen et al. (2023), and concurrently kaiokendev (2023). Under the notation of Eq. 7, it is setting

$$g(m) = s \cdot m, \mathbf{h}(\theta) = \theta, \quad (9)$$

where s is the scale factor $\frac{L'}{L}$. We include some details in Appendix A.1.

ReRoPE (Su, 2023) also aims to extend the context size of existing models pre-trained with RoPE, and claims "infinite" context length without needing any fine-tuning. This claim is backed by a monotonically decreasing loss with increasing context length up to 16k on the Llama 2 13B model. It achieves context extension by modifying the attention mechanism and thus is not purely an embedding interpolation method. Since it is currently not compatible with Flash Attention 2 (Dao, 2023) and requires two attention passes during inference, we do not consider it for comparison.

Concurrently with our work, LM-Infinite (Han et al., 2023) proposes similar ideas to YaRN, but focuses on "on-the-fly" length generalization for non-fine-tuned models. Since they also modify the attention mechanism of the models, it is not an embedding interpolation method and is not immediately compatible with Flash Attention 2.

3 METHODOLOGY

Whereas PI stretches all RoPE dimensions equally, we find that the theoretical interpolation bound described by PI (Chen et al., 2023) is insufficient at predicting the complex dynamics between RoPE and the LLM’s internal embeddings. In the following subsections, we describe the main issues with PI we have individually identified and solved, so as to give the readers the context, origin and justifications of each method which we use in concert to obtain the full YaRN method.

3.1 LOSS OF HIGH FREQUENCY INFORMATION - "NTK-AWARE" INTERPOLATION

If we look at rotary position embeddings (RoPE) only from an information encoding perspective, it was shown in (Tancik et al., 2020), using Neural Tangent Kernel (NTK) theory, that deep neural networks have trouble learning high frequency information if the input dimension is low and the corresponding embeddings lack high frequency components. Here we can see the similarities: a token’s positional information is one-dimensional, and RoPE expands it to an n -dimensional complex vector embedding. RoPE closely resembles Fourier Features (Tancik et al., 2020) in many aspects, as it is possible to define RoPE as a special 1D case of a Fourier Feature.

In the case of Positional Interpolation (PI), as we stretch all dimensions equally by a factor s , it removes the high frequency components of RoPE. This degradation is worsened as the scaling factor s grows, and at some point, the network will not be able to recover. Previous fine-tunes (kaiokendev, 2023) (Chen et al., 2023) (Together.ai, 2023) (Quesnelle et al., 2023) using PI were only able to achieve a scaling factor of roughly $s = 8$ before the LLM’s outputs starts to degrade, even after fine-tuning.

In order to alleviate this issue, the "NTK-aware" interpolation was developed in (bloc97, 2023a). Instead of scaling every dimension of RoPE equally by a factor s , we spread out the interpolation pressure across multiple dimensions by scaling high frequencies less and low frequencies more. One can obtain such a transformation in many ways, but the simplest would be to perform a base change on the value of θ . The details are described in the Appendix A.2 and the method has seen some open-source adoptions*.

One main issue of this "NTK-aware" scaling is that it is very difficult to determine what optimal base should be used for an intended context extension by s times. The best base to use for "NTK-aware" interpolation usually has to be found empirically, which significantly increases the difficulty and cost of obtaining a successful fine-tuned model. Despite its limitations, the observations from the NTK theory is valid and the following idea is still maintained and executed in a different way in the "NTK-by-parts" interpolation introduced in the next section.

3.2 LOSS OF RELATIVE LOCAL DISTANCES - "NTK-BY-PARTS" INTERPOLATION

To understand why "NTK-aware" interpolation works better than PI and to eliminate its disadvantages, we have to take a closer look at RoPE. In this section, we think heavily in terms of the wavelengths λ_d defined in Eq. 8 in the formula of RoPE. For simplicity, we omit the subscript d in λ_d and the reader is encouraged to think about λ as the wavelength of an arbitrary periodic function.

In theory, as RoPE is a relative position embedding, it should be quite surprising that it fails to generalize to unseen longer context sizes. However, we can show that in practice, RoPE does not only encode relative position. One observation we can make is that given a context size L , there are some dimensions d where the wavelength is longer than the maximum context length seen during pretraining ($\lambda > L$), this suggests that some dimensions’ rotary embeddings might not be distributed evenly in the rotational domain (i.e. does not perform a full rotation for the entire training context size). In such cases, we presume having unique position pairs[†] implies that the absolute positional

*We note that shortly before the release of this article, Code Llama (Rozière et al., 2023) was released and uses "NTK-aware" scaling by manually scaling the base b to 1M, in which they call this method as RoPE "adjusted base frequency" (ABF).

[†]Since the dimension never rotates fully at least once during pre-training, if we pick the first token as the anchor, every other token during pre-training has a unique distance to it, which the neural network can use to determine its absolute position.

information remains intact in those dimensions. On the contrary, when the wavelength is short, only relative positional information is accessible to the network.

Given these observations, we can see that it is important to not touch the dimensions that only encode relative positional information, as they are crucial for the network to distinguish the relative order of nearby tokens. Meanwhile, dimensions that only encode absolute positional information should always be interpolated, as larger distances will be out of distribution. Instead of arbitrarily changing the base in "NTK-aware" interpolation (which basically does something similar to what is described here), we can formulate an explicit and targeted interpolation method that takes in account all of the above.

In other words,

- if the wavelength λ is much smaller than the context size L , we do not interpolate;
- if the wavelength λ is equal to or bigger than the context size L , we want to only interpolate and avoid any extrapolation (unlike the previous "NTK-aware" method);
- dimensions in-between can have a bit of both, similar to the "NTK-aware" interpolation.

As a result, it is more convenient to introduce the ratio $r = \frac{L}{\lambda}$ between the original context size L and the wavelength λ . This ratio represents the number of rotations a certain RoPE dimension makes given a fixed pretrained context length L . In the d -th hidden state, the ratio r depends on d in the following way:

$$r(d) = \frac{L}{\lambda_d} = \frac{L}{2\pi b^{\frac{2d}{|D|}}}. \quad (10)$$

In order to define the boundary of the different interpolation strategies as above, we introduce two extra parameters α, β . All hidden dimensions d where $r(d) < \alpha$ are those where we linearly interpolate by a scale s (exactly like PI, avoiding any extrapolation), and the d where $r(d) > \beta$ are those where we do not interpolate at all. Define the ramp function γ to be

$$\gamma(r) = \begin{cases} 0, & \text{if } r < \alpha \\ 1, & \text{if } r > \beta \\ \frac{r - \alpha}{\beta - \alpha}, & \text{otherwise.} \end{cases} \quad (11)$$

With the help of the ramp function, the "NTK-by-parts" method can be described as follows.

Definition 1 *The "NTK-by-parts" interpolation is a modification of RoPE using Eq. 7 with the following functions[‡].*

$$g(m) = m \quad (12)$$

$$h(\theta_d) = \left(1 - \gamma(r(d))\right) \frac{\theta_d}{s} + \gamma(r(d)) \theta_d. \quad (13)$$

The values of α and β should be tuned on a case-by-case basis. For example, we have found experimentally that for the Llama family of models, good values for α and β are $\alpha = 1$ and $\beta = 32$.

Using the techniques described in this section, a variant of the resulting method was released under the name "NTK-by-parts" interpolation (bloc97, 2023b). This improved method performs better than the previous PI (Chen et al., 2023) and "NTK-aware" 3.1 interpolation methods, both with non-fine-tuned models and with fine-tuned models, as shown in (bloc97, 2023b) and Section 4.2.

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In addition to the previous interpolation techniques, we also observe that introducing a temperature t on the logits before the attention softmax has a uniform impact on perplexity regardless of the data

[‡]The interpolation by linear ramp on h may have alternatives, such as a harmonic mean over θ_d/s and θ_d converted from a linear interpolation on wavelengths. The choice of h here was for the simplicity of implementation, but both would work.